

Double-Buffered SIGReg

Working Paper

Aurélien Cecille

May 13, 2026

Abstract

Self-supervised learning (SSL) architectures that rely on distribution matching, such as those utilizing the Epps-Pulley empirical characteristic function test, enforce regularization by shaping the latent space into an isotropic Gaussian. A major bottleneck of these approaches is the requirement for massive batch sizes to accurately estimate the collective distribution, leading to heavy memory and computational constraints. While temporal aggregation (e.g., Exponential Moving Averages) of statistics could theoretically decouple batch size from distribution estimation, it fails in practice due to the necessity of continuously resampling random projection matrices to avoid dimensional collapse in high-dimensional spaces (the “Null Space Trap”). In this paper, we propose a Double-Buffered Random Projection mechanism. By maintaining an active projection buffer for gradient computation and a shadow buffer for silent statistical warm-up, our method allows for temporal aggregation of characteristic functions without falling into dimensional blind spots or suffering from gradient shock. This architecture achieves the regularization stability of massive batch sizes at a fraction of the compute and memory overhead.

1 Introduction

Modern Representation Learning heavily relies on preventing representation collapse. While contrastive methods explicitly repel negative samples, regularization-based methods enforce structural properties on the latent space. Recent advances, such as Joint-Embedding Predictive Architectures regularized via the Epps-Pulley test, force the representation manifold to match a target isotropic Gaussian distribution.

A fundamental limitation of matching distributions is that a distribution is a collective property of a batch, not an individual sample. To prevent highly volatile gradient updates driven by noisy empirical means, these architectures require exceptionally large batch sizes (e.g., $B \geq 4096$). This imposes a massive memory footprint, as the intermediate activations of the backbone encoder (typically a Vision Transformer) scale quadratically with sequence length and linearly with batch size.

To alleviate this, one might naturally attempt to aggregate the distribution statistics over time using an Exponential Moving Average (EMA). However, computing the distribution distance in high-dimensional Euclidean space is intractable, necessitating the projection of embeddings onto random 1D axes. The inherent contradiction arises here: to prevent the network from hiding collapsed representations in the null space of the projections, the random axes must be frequently re-sampled. Yet, if the axes change, the historical EMA of the distribution becomes instantly invalid.

We introduce **Double-Buffered Random Projections**, a mechanism inspired by computer graphics and systems engineering, to resolve this contradiction. By maintaining two overlapping states—an active projection matrix providing stable gradients and a shadow projection matrix accumulating valid statistics in the background—we seamlessly rotate the geometric evaluation frame without invalidating temporal aggregates.

2 Preliminaries

2.1 The Characteristic Function Objective

The Epps-Pulley test operates in the frequency domain, measuring the distance between the Empirical Characteristic Function (ECF) of the batch and the theoretical characteristic function of a standard normal distribution, $\phi(t) = \exp(-t^2/2)$. For a set of 1D scalar projections z , the test evaluates:

$$\mathcal{L}_{EP} = \sum_{k=1}^T w_k \left| \frac{1}{B} \sum_{j=1}^B e^{it_k z_j} - e^{-t_k^2/2} \right|^2 \quad (1)$$

where t_k and w_k are fixed quadrature points and weights derived from Gauss-Hermite integration. The loss naturally decouples into cosine and sine terms, allowing for independent summation across the batch.

2.2 The Random Projection Contradiction

To apply this to a D -dimensional latent space, the representations $Z \in \mathbb{R}^{B \times D}$ are projected onto a matrix of A random directions, $P \in \mathbb{R}^{D \times A}$, yielding $Z_{proj} = ZP$. According to the Cramér-Wold theorem, Z is Gaussian if and only if all its 1D projections are Gaussian.

- **If P is fixed forever:** The neural network learns to map unregularized, collapsed structures into the null space of P . The 1D projections appear perfectly Gaussian, but the D -dimensional space is severely distorted.
- **If P is re-sampled at every step:** We prevent the null space trap, but we cannot aggregate $\cos(t \cdot ZP)$ over time, as the reference frame P constantly shifts, forcing us to rely on the current batch size B for the ECF estimate.

3 Method: Double-Buffered Projections

To combine the memory efficiency of temporal aggregation with the geometric robustness of moving projections, we decouple the gradient generation step from the statistical aggregation step.

3.1 The Active and Shadow Buffers

We define a window size of K steps. The system initializes two identical-sized random projection matrices:

1. A_{active} : The active projection matrix.
2. A_{shadow} : The shadow projection matrix.

During steps 1 to K , the network executes the forward pass through both matrices:

$$Z_{active} = ZA_{active} \quad (2)$$

$$Z_{shadow} = ZA_{shadow} \quad (3)$$

3.2 Step 1: Gradient Computation via Active Buffer

The loss and gradients are computed strictly using Z_{active} and its mature, temporally aggregated state. Let $\mathcal{H}_{\cos}(t)$ represent the historical mean of the cosine frequencies. The active error is computed as:

$$Error_{active}(t_k) = \left(\mathcal{H}_{\cos}^{(active)}(t_k) - \exp(-t_k^2/2) \right)^2 + \left(\mathcal{H}_{\sin}^{(active)}(t_k) \right)^2 \quad (4)$$

The gradient ∇Z is derived from this error and backpropagated through A_{active} to update the network weights.

3.3 Step 2: Silent Warm-up via Shadow Buffer

Simultaneously, the batch Z_{shadow} is used to compute empirical frequencies for the newly instantiated random directions in A_{shadow} . *Crucially, this operation requires no gradients (requires_grad=False).* We update the shadow history:

$$\mathcal{H}_{\cos}^{(shadow)}(t_k) \leftarrow \text{Update} \left(\mathcal{H}_{\cos}^{(shadow)}(t_k), \frac{1}{B} \sum_{j=1}^B \cos(t_k \cdot Z_{shadow,j}) \right) \quad (5)$$

3.4 The Update Rule: CMA vs. EMA

The Update function can be instantiated as either a Cumulative Moving Average (CMA) or an Exponential Moving Average (EMA).

- **CMA:** Perfect statistical weighting over exactly K steps. Ideal for environments with very low learning rates where representation drift is negligible.
- **EMA:** Weights recent samples higher, serving as an implicit forgetting mechanism for out-dated network representations caused by weight drift during the K -step window.

3.5 Step 3: The Swap Mechanism

Upon reaching step K , the active buffer has successfully prevented dimensional collapse along its axes, but the network may be beginning to exploit its null space. At this exact step:

1. A_{active} is discarded.
2. A_{shadow} (and its fully matured statistical history $\mathcal{H}$) is promoted to become the new A_{active} .
3. A new random matrix is generated to become the new A_{shadow} , with its history initialized to zero.

This swap is mathematically seamless. The newly promoted active buffer has a statistically valid, un-noised historical average computed over the exact same target distribution, preventing any gradient shock or loss spikes during the rotation of the projection axes.

4 Computational Complexity

The computational overhead of the shadow buffer is negligible. Given a latent dimension $D = 8192$ and $A = 1024$ projection directions, the addition of the shadow buffer requires a single matrix concatenation resulting in a $(B \times D) \times (D \times 2A)$ multiplication. On modern hardware accelerators, the cost of widening the terminal linear projection is virtually zero, while allowing the simulated batch size to effectively scale from B to $N = K \times B$, vastly reducing the required activation memory in the Transformer backbone.

5 Conclusion and Future Work

We have outlined a theoretical framework that bridges the gap between massive-batch regularization algorithms and the physical constraints of single-GPU/consumer hardware. By decoupling the projection axes using a double-buffered shadow update, we allow temporal distribution matching in high-dimensional spaces without succumbing to null space vulnerabilities.

[Experimental section and empirical validation to be added.]